

Center of Gravity

Definition: Center of gravity of an object is the imaginary point where the entire weight of the object seems to act

#PhysicsDefinitions

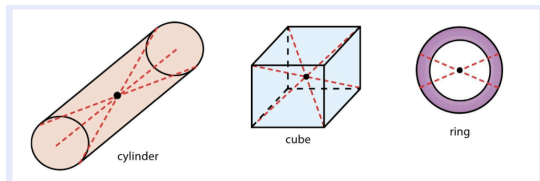
For object to be in balance $\Rightarrow$ sum of clockwise moments due to individual particles must be equal to sum of anticlockwise moments.

$\Rightarrow$ equivalent to having force acting through a point below pivot, since net turning effect is 0.

$\Rightarrow$ can also represent sum of all individual particles weight by a resultant force acting through this point.

Resultant force is the weight of the body

Center of gravity of an object $\Rightarrow$ that has regular shape and density $\Rightarrow$ is at centroid or centre of object



NOTE: Center of gravity does not need to lie within the object. example is with ring

Finding Centre of Gravity of Irregularly Shaped Object

MEMORISE*

1. Make small holes near edges of the lamina
2. holes should be as far away as possible
3. suspend lamina freely from a pin. lamina will come to rest when centre of gravity is below the pin.
4. Hang a plumb line from pin in front of lamina
5. mark top and bottom of plumb line on lamina
6. Repeat for the other holes
7. Place lamina on the table and join the markings

Centre of gravity $\Rightarrow$ is point of intersection of lines you have drawn before.

(for some irregularly shaped lamina $\Rightarrow$ point of intersection lies outside Lamina Itself)

Precautions

- switch off fans $\Rightarrow$ wind may affect equilibrium position of lamina
- use a thin thread for plumb line
- ensure plumb line is not swinging when marking the point
- ensure that the holes on the lamina are small but still allow lamina to swing freely. Frictional force between pin and lamina may affect equilibrium position